What Do Benign-Trained Session Language Models Actually Detect? A Mechanistic Audit on Insider-Threat Benchmarks

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Paper under double-blind review

Abstract

Score-level evaluation cannot reveal *what* an anomaly detector has learned to detect. We develop a mechanistic audit for anomaly language models—sparse-autoencoder decomposition of adapter-induced representation changes, validated by causal interventions against matched controls and labeled by token-level attribution—and apply it to Qwen3-8B adapted with QLoRA on benign users only, across two CERT insider-threat benchmarks. Despite similar scores, the audited detectors rest on qualitatively different internal evidence. On r6.2, whose few malicious users are profile-distinctive within the benign population, the causally implicated features are almost entirely profile-bound (99.8–100% of activation mass on simulated psychometric and organizational tokens), consistent with user-profile novelty; on r4.2, four of five causally selected features instead concentrate on behavioral session events; and subsampling shows that raw positive count alone does not explain the difference. What reproduces is the pattern, not the individual features: literal feature transfer across benchmarks fails, but each benchmark's profile-versus-behavior evidence type recurs across SAE initializations and benign-only dictionaries, and in smaller reruns with a repaired serializer and fresh adapter seeds. The resulting account—when malicious users are systematically distinguishable from the benign adaptation population through profile attributes, profile novelty offers a cheap shortcut to sequence-likelihood detectors (classical detectors on the same features do not take it); when that shortcut is unavailable, behavioral evidence dominates—was specified on CERT and makes an out-of-setting prediction: on TWOS, with real users, real psychometrics, and no profile-distinctive malicious minority, the audit should certify behavioral evidence, and it does. The score-level consequence is stark: compared only against benign users unseen during adaptation, day-level ROC-AUC falls from 0.95 to 0.55 (r6.2) and from 0.96 to 0.67 (r4.2)—apparent strength is largely a seen-versus-unseen-user effect—and score-time profile masking *improves* r4.2's unseen-user discrimination. High anomaly-ranking performance can thus conceal qualitatively different—and sometimes identity-proxy—internal evidence; mechanistic auditing makes the difference visible.

1 Introduction

Enterprise anomaly detection is usually evaluated as a ranking problem: a model assigns scores, and the question is whether malicious activity ranks above benign activity. In high-stakes settings such as insider-threat monitoring, a score alone answers the wrong question. Two detectors with identical ROC curves can rest on entirely different evidence—one on anomalous *behavior*, another on incidental properties of who the malicious users happen to be—and no amount of score-level evaluation can tell them apart (Chandola et al., 2009; Yuan & Wu, 2021). Mechanistic interpretability offers the missing instrument: if the internal computation that raises an anomaly score can be localized to sparse, causally validated features, then what the detector actually detects becomes an empirical question (Olah et al., 2020; Bricken et al., 2023).

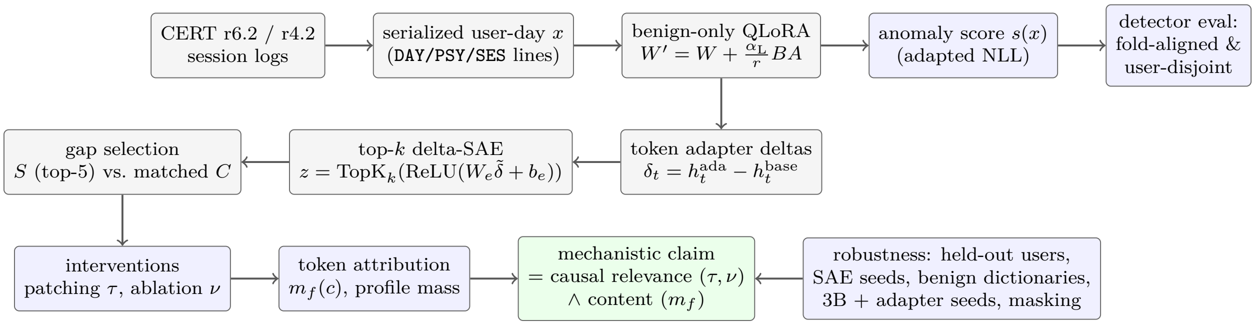

Figure 1: The audit pipeline, annotated with the notation of Section 4.1. Top row (model and score): session logs are serialized into structured user-day text x and adapt Qwen3-8B on benign users only; the adapted NLL $s(x)$ is evaluated under both the fold-aligned and the user-disjoint detector protocols. Middle row (representation): token-level adapter deltas δ_t feed a top-k sparse autoencoder, and features are selected by activation gap against matched active controls. Bottom row (audit): intervention estimands (τ: patching; ν: ablation) certify causal relevance, token attribution $m_f(c)$ identifies content, and only their conjunction—stress-tested by the robustness battery—licenses a mechanistic claim.

This paper develops that instrument as a *mechanistic audit* and reports what it finds on the standard insider-threat benchmarks. We adapt Qwen3-8B (Qwen Team, 2025) with QLoRA (Hu et al., 2022; Dettmers et al., 2023) on structured session logs from benign users only, isolate the adapter's token-level effect as residual-stream deltas, decompose those deltas with top-k sparse autoencoders (Gao et al., 2024), and subject selected feature sets to causal patching and necessity ablation against matched active controls (Zhang & Nanda, 2023). Because intervention effects on observational populations are easily inflated, the audit carries its own validity program: user-level cluster uncertainty, same-user donor exclusion, held-out-user discovery/confirmation splits, and SAE-seed stability controls. Finally—and decisively—token-level attribution asks where the causally implicated features actually fire in the serialized input.

The audit's findings are a dissociation. On CERT r6.2, whose evaluation population contains four malicious users, the causally implicated features are almost entirely *profile-bound*, placing essentially all of their activation mass on static psychometric and organizational tokens rather than behavioral session events, consistent with user-profile novelty. On CERT r4.2, whose population contains sixty malicious users, the natively rediscovered mechanism is *behavioral*: four of five top features concentrate on session-event tokens such as durations. Both mechanisms are causally relevant under our estimands—r4.2's replicates in direction on held-out users (feature-level independence; one context individually significant), while r6.2's held-out replication concentrates on its dominant user—and each feature family is stable across SAE initializations with no counterpart in the other benchmark's dictionary. What differs is what the selected features encode. Because the session model is adapted on benign users only, malicious-population structure cannot alter adapter training; it can, however, shape both the SAE dictionary (which is fitted on the evaluation pool) and which directions the audit selects. We tested the natural selection-stage hypothesis—small positive populations favor profile-bound selection—by subsampling r4.2's malicious users within its fixed dictionary, and found it insufficient: even four-user subsamples select predominantly behavioral features. Retraining both dictionaries on benign rows alone reproduces the attribution dissociation, so profile capture does not require positive-shaped dictionaries either. r6.2's profile capture is a property of that benchmark's label–profile structure and its adapter's learned geometry, not of small positive populations in general. These conclusions extend beyond the two synthetic benchmarks: the behavioral pole reproduces on a real-user dataset with genuine psychometrics (Section 6.8), and classical detectors given the same profile fields do not take the shortcut, locating it in the sequence-likelihood pathway (Section 7).

Contributions.

1. We assemble an intervention-based *mechanistic audit* for anomaly language models—adapter-delta sparse autoencoders, causal patching and necessity ablation under active controls, cluster-level uncer-

tainty, dictionary-conditional held-out-user confirmation, seed-stability controls, and token-level feature attribution—and apply and stress-test the combined audit on two CERT benchmarks.

2. Using the audit, we show that the same architecture and pipeline learns mechanisms of different *kinds* on related benchmarks: profile-bound features consistent with user-profile novelty on r6.2 (99.8–100% of top-feature activation mass on psychometric and organizational tokens) versus behavioral session features on r4.2—a failure mode invisible to score-level evaluation. The dissociation is established for these benchmark–adapter instances (one adaptation run per benchmark); population-level tests reject positive-count explanations at the selection stage, and benign-only dictionary retraining rejects dictionary-formation explanations.

3. We show the two mechanisms are stable subspaces of their own models (cross-seed decoder alignment 0.81–0.95 in residual-stream coordinates, above dictionary-median null baselines of 0.72–0.82) while cross-benchmark top-feature alignment sits at or below its null—at matched depth, the top features align *worse* than typical dictionary features—showing that SAE initialization instability is unlikely to explain the observed transfer failure.

4. We test the account's generality on real data and across detector classes: on TWOS (real users, real Big-Five psychometrics) the audit certifies behavioral evidence exactly where the account predicts it, while classical one-class detectors given the same CERT profile fields do not take the profile shortcut, locating it in the sequence-likelihood pathway. The resulting two-factor account—population structure governs when an identity shortcut is available; objective and representation govern which detectors take it—is stated with its isolating factorial tests named as future work.

5. We document the audit's validity program: why receiver-level uncertainty, permissive donor matching, and search-then-evaluate selection each overstate mechanistic evidence, and how cluster bootstraps, same-user exclusion, and feature-level discovery/confirmation splits repair them.

2 Related Work

Insider-threat detection. The CERT insider-threat test datasets (Glasser & Lindauer, 2013; Lindauer, 2020) are the standard public benchmarks for user-behavior-based insider-threat detection. Prior work spans unsupervised deep models over aggregated user-day features (Tuor et al., 2017), systematic studies of data granularity (Le et al., 2020), and a broad deep-learning literature reviewed by Yuan & Wu (2021). Our baseline detectors follow this tradition: one-class detectors (Deep SVDD (Ruff et al., 2018), GRU and LSTM autoencoders, Isolation Forest (Liu et al., 2008)) over session-level behavioral features. Our contribution is orthogonal to detector engineering: we ask whether a sequence-native language model trained on the same benign population admits a causal, feature-level account of its anomaly evidence.

Language models over logs. Treating system and behavior logs as sequences for anomaly detection predates modern LLMs (Du et al., 2017; Guo et al., 2021). These approaches inherit the transformer's sequence modeling capacity (Vaswani et al., 2017) but are typically evaluated only as detectors. We instead use parameter-efficient adaptation (Hu et al., 2022; Dettmers et al., 2023) of a general-purpose LLM (Qwen Team, 2025) and center the evaluation on the internal structure induced by adaptation rather than on detection leaderboards. This detector class is no longer hypothetical: LoRA-tuned and LLM-based insider-threat log analysis systems are actively being developed (Kong et al., 2025; Li et al., 2025), which makes auditing what such detectors learn a timely obligation rather than a retrospective one; to our knowledge, ours is the first end-to-end causal audit of what an adaptation-based anomaly detector's score rests on.

Mechanistic interpretability. Sparse autoencoders and dictionary learning decompose model activations into sparse, more interpretable features (Cunningham et al., 2023; Bricken et al., 2023; Gao et al., 2024). Intervention-based validation—activation patching, causal mediation, and ablation—provides the causal counterpart (Vig et al., 2020; Meng et al., 2022; Zhang & Nanda, 2023; Heimersheim & Nanda, 2024), with causal scrubbing formalizing hypothesis testing over internal computations (Chan et al., 2022).

Circuit-level studies have largely targeted curated natural-language behaviors (Wang et al., 2023). Two design choices in our protocol follow explicit recommendations from this literature: effects are always measured against matched active controls rather than in isolation (Zhang & Nanda, 2023), and we treat cross-dataset transfer of a mechanism as an empirical claim to be tested rather than assumed, connecting to the question of universality of learned structure (Olah et al., 2020; Chughtai et al., 2023). Like recent work analyzing LoRA-induced *delta activations* with sparse autoencoders (Prasanth, 2026), our decomposition targets what fine-tuning adds to the residual stream rather than raw activations; our contribution is not the delta-SAE construction itself but its combination with intervention-based validation under matched controls, user-disjoint security evaluation, and semantic attribution into an end-to-end audit.

Shortcut learning and evaluation pitfalls. Our central finding is an instance of shortcut learning: models satisfying a training objective through features that do not generalize as intended (Geirhos et al., 2020; Lapuschkin et al., 2019), a risk amplified when pipelines are underspecified by their evaluations (D'Amour et al., 2022). Closest to our setting, the Clever Hans effect has been demonstrated in anomaly detection specifically, with explanation methods revealing detectors that rely on spurious input components (Kauffmann et al., 2020); our audit extends that program from attribution over inputs to causally validated sparse features inside an adapted language model. In security applications specifically, sampling bias, label artifacts, and inappropriate train/test separation are documented recurrent pitfalls (Arp et al., 2022); our user-disjoint detector analysis and profile-masking counterfactuals are instances of the evaluation hygiene that work prescribes. On the interpretability side, SAE feature identity is itself an unstable object—features can split or absorb one another across dictionary configurations (Chanin et al., 2024)—which motivates our seed controls, matched-layer alignment baselines, and the benign-only dictionary retraining check.

3 Benchmarks and Session Modeling

Datasets. We use CERT r6.2 and r4.2 (Glasser & Lindauer, 2013; Lindauer, 2020), simulated enterprise environments with ground-truth insider incidents. Following the LC-DAL feature-extraction line used by our baseline detectors, raw event logs are aggregated into per-session behavioral feature vectors (logon/logoff structure, file, device, HTTP, and email activity), grouped into user-days. Each example in our corpus is one user-day.

Evaluation population. All results are stated over the *matched active-session* domain: user-days that appear in the LC-DAL session-feature extraction. This domain is smaller than the raw answer key, because some labeled incidents fall outside active session coverage. For r6.2, the matched domain contains 70 positive user-days across 4 malicious users (the answer key lists 99 positive rows across 5 users); for r4.2, it contains 1309 positive user-days across 60 users (answer key: 1883 rows across 70 users). We report this explicitly because population re-filtering is a common source of silent optimism in insider-threat evaluation.

Serialization. Each user-day is rendered as structured text: a `DAY` header with organizational context (week, project, role, business and functional unit, department, team, IT-administrator flag), a `PSY` line with the five simulated psychometric traits, a session-count line, and one `SES` line per session listing its numeric LC-DAL features as `key=value` pairs. The serialization is deliberately non-prose: the model must learn behavioral regularities from structured key-value sequences, not natural-language cues. Four hyphenated file-transfer columns are dropped by the serializer due to a column-naming constraint (all four exist in r6.2's raw schema; r4.2's release contains only one of them); we treat the serialized field set, not the full LC-DAL schema, as the model's observation space. A smaller-scale rerun with the defect repaired reproduces the paper's central attribution result (Section 6.6).

Splits. Users with any positive day are assigned entirely to an evaluation split; benign users are hashed to a validation split with probability 0.10 and otherwise to training. Training and early stopping therefore never see malicious users, making the adapted model a one-class detector at the population level, not merely at the label level.

4 Model and Sparse Feature Extraction

Benign-only adaptation. We adapt Qwen3-8B (Qwen Team, 2025) with QLoRA (Dettmers et al., 2023): the base model is quantized to 4-bit NF4 with double quantization and bfloat16 compute, and low-rank adapters (Hu et al., 2022) with rank 16, scaling $\alpha = 32$, and dropout 0.05 are attached to all attention and MLP projection matrices. Training minimizes next-token negative log-likelihood on benign training user-days only (sequence length 2048, effective batch size 8, one epoch, learning rate 1.5×10^{-4} with cosine decay and 3% warmup). The per-example anomaly score is the length-normalized NLL of the serialized user-day under the adapted model (*adapted NLL*); we also report base-model NLL and their difference as reference scores.

Token-level adapter deltas. For a layer ℓ, let h_t^{ada} and h_t^{base} be the residual-stream states of token t under the adapted and base model on the same input. The *adapter delta* $\delta_t = h_t^{\text{ada}} - h_t^{\text{base}}$ isolates what benign-only adaptation changed about the representation of token t. We train top-k sparse autoencoders (Gao et al., 2024) on token deltas from the evaluation pool—all positive-example rows plus subsampled benign rows (Appendix A)—using a ReLU encoder, a top-k mask, and an untied linear decoder, with latent dimension a multiplier m of the model width. Because the dictionary is fitted on the full evaluation pool, all held-out analyses below provide feature-selection and evaluation independence *conditional on this shared dictionary*, not dictionary-level independence; we flag this wherever held-out results are claimed. As a dictionary-independence check, we additionally retrained both SAEs from scratch on benign rows only—no positive example enters dictionary formation—and reran selection, attribution, and held-out evaluation (Section 6.6): the attribution dissociation reproduces exactly, while replication of the intervention estimands is partly dictionary-sensitive. The headline r6.2 configuration uses layer 18, $m = 4$, $k = 8$; the headline r4.2 configuration, found by native search (Section 6.3), uses layer 26, $m = 2$, $k = 4$. Candidate feature sets are the top-5 features ranked by association between feature activity and positive examples.

4.1 Formal pipeline

We collect the full pipeline—adaptation, scoring, delta extraction, sparse decomposition, intervention estimands, and attribution—in one place.

Benign-only QLoRA. Let p_{θ_0} denote the frozen base model with NF4-quantized weights. For each adapted linear map $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, QLoRA parameterizes $W' = W + \frac{\alpha_{\text{L}}}{r} BA$ with $B \in \mathbb{R}^{d_{\text{out}} \times r}$, $A \in \mathbb{R}^{r \times d_{\text{in}}}$ (r=16, α_{L}=32), and only (A, B) are trained on the benign corpus $\mathcal{D}_{\text{ben}}$ of serialized user-days:

$$\mathcal{L}(\theta) \;=\; -\sum_{x \in \mathcal{D}_{\text{ben}}} \sum_{t=1}^{|x|} \log p_\theta(x_t \mid x_{<t})\,. \tag{1}$$

No positive example influences θ.

Anomaly score. For a serialized day x, the detector score is the length-normalized adapted negative log-likelihood

$$s(x) \;=\; -\tfrac{1}{|x|} \textstyle\sum_t \log p_{\theta_{\text{ada}}}(x_t \mid x_{<t})\,, \tag{2}$$

with s_{base} defined analogously under θ_0; days are ranked by s.

Adapter deltas. Let $h_t^{(\ell)}(x; \theta) \in \mathbb{R}^d$ be the layer-ℓ residual-stream state at position t. The token-level adapter delta

$$\delta_t(x) \;=\; h_t^{(\ell)}(x; \theta_{\text{ada}}) - h_t^{(\ell)}(x; \theta_0) \tag{3}$$

isolates what adaptation adds to the computation at (ℓ, t); it is the object we decompose.

Top-k sparse autoencoder. Deltas are standardized per coordinate, $\tilde{\delta} = (\delta - \mu) \oslash \sigma$, and encoded by

$$z(\tilde{\delta}) \;=\; \text{TopK}_k\big(\text{ReLU}(W_e \tilde{\delta} + b_e)\big) \in \mathbb{R}^{md}_{\geq 0}, \qquad \hat{\delta} \;=\; W_d\, z, \tag{4}$$

where TopK_k zeroes all but the k largest pre-activations and $W_d \in \mathbb{R}^{d \times md}$ is an untied linear decoder trained to minimize $\mathbb{E}\, \|\tilde{\delta} - \hat{\delta}\|_2^2$. The residual-stream direction of feature f is $u_f = \sigma \odot W_d[:, f]$ (all cross-dictionary geometry is computed on normalized u_f, not on raw decoder columns, which live in dictionary-specific standardized coordinates).

Feature selection. With row labels y inherited from examples, the selection statistic is the activation gap $g_f = \mathbb{E}[z_f \mid y{=}1] - \mathbb{E}[z_f \mid y{=}0]$; the audited set S is the top-5 by g_f and C is a matched active control set ($|C| = |S|$, activation frequency ≥ 0.002, low $|g_f|$; dose-matched variants additionally match activation magnitude and decoder norm).

Causal patching (sufficiency-style). For receiver x with codes $\{z_t\}$, feature set F, donor day d with prototype p_f (mean donor code over the donor's F-active positions), and strength $\alpha \in \{0.25, 0.5, 0.75, 1\}$, the counterfactual codes are

$$z'_{t,f} \;=\; (1-\alpha)\, z_{t,f} + \alpha\, p_f \quad \text{for } f \in F,\ t \in T_F, \qquad T_F = \{t : \textstyle\sum_{f\in F} z_{t,f} > 0\}, \tag{5}$$

and the injected residual shift at every token is $\Delta_t = \big(\sigma \odot W_d z'_t + \mu\big) - \delta_t$, applied by forward hook at layer ℓ during scoring. Writing $s_F(x; d, \alpha)$ for the patched score, the recorded per-receiver outcome under donor class $c \in \{\text{ben}, \text{anom}\}$ is the maximal achievable repair $\Delta_c^F(x) = \min_{d\in\mathcal{D}_c(x),\, \alpha}\big[s_F(x; d, \alpha) - s(x)\big]$, and the estimand is the paired difference-in-differences

$$\tau \;=\; \underbrace{\big(\overline{\Delta^S_{\text{anom}}} - \overline{\Delta^S_{\text{ben}}}\big)}_{\text{RA}(S)} \;-\; \underbrace{\big(\overline{\Delta^C_{\text{anom}}} - \overline{\Delta^C_{\text{ben}}}\big)}_{\text{RA}(C)}, \tag{6}$$

computed over complete-case receivers with same-user donors excluded; uncertainty is a user-level cluster bootstrap over malicious users.

Necessity ablation. Identical machinery with $z'_{t,f} = (1-\alpha) z_{t,f}$ (no donor); with $e_F(x)$ the resulting score change, the estimand contrasts ablation dependence across receiver classes and feature sets, $\nu = \big[\overline{e_S(x_{\text{ben}})} - \overline{e_S(x_{\text{pos}})}\big] - \big[\overline{e_C(x_{\text{ben}})} - \overline{e_C(x_{\text{pos}})}\big]$ over context-matched pairs.

Token attribution. Each position carries a line class $c(t) \in \{\texttt{DAY}, \texttt{PSY}, \texttt{SESCOUNT}, \texttt{SES}\}$ from the serialization schema. The attribution mass of feature f on class c over positive rows is

$$m_f(c) \;=\; \frac{\sum_{(x,t)\,:\,c(t)=c} z_{t,f}(x)}{\sum_{(x,t)} z_{t,f}(x)}, \qquad \text{profile mass} \;=\; m_f(\texttt{DAY}) + m_f(\texttt{PSY}). \tag{7}$$

From interventions to mechanism. The audit's inferential contract is the conjunction: $\tau > 0$ and $\nu > 0$ against matched controls establish that S is *causally implicated* in the score's anomaly evidence (with the estimand-specific caveats of Appendix A); the attribution profile $\{m_f\}_{f\in S}$ establishes *what* S encodes; and only together do they license mechanistic statements of the form "the detector's anomaly evidence rides on profile (or behavioral) content." Either component alone underdetermines the mechanism: interventions certify causal relevance of an unlabeled subspace, and attribution labels a subspace without certifying relevance.

5 Evaluation Protocol

5.1 Fold-aligned detector benchmark

The adapted session LM and the feature-based baselines are compared under an identical fold construction: one held-out malicious user per fold, 800 sampled benign test users per fold (fixed seed), day-level metrics over all test-user rows, and user-level metrics after max-aggregation over days. The adapted model is trained once on the benign population and held fixed across folds; the feature-based baselines retrain per fold on non-test benign users. This asymmetry matters and we state it plainly: the session LM's training corpus covers roughly 90% of benign users, so most sampled benign test users were seen (as benign sequences) during adaptation, whereas the baselines exclude test users from training. The bias is not hypothetical: restricting the benign comparison population to never-trained validation-split users collapses day-level ROC-AUC from 0.954 to 0.545 on r6.2 and from 0.959 to 0.668 on r4.2 (user-level: $0.970 \to 0.546$ and $0.969 \to 0.565$). These paired values are computed on the full evaluation pool; the fold-aligned values in Table 1 (0.953/0.964) use the baselines' per-fold test populations and therefore differ in the last digits. Since malicious users are unseen by construction, the unseen-versus-unseen comparison is the fairer test of generalization to unseen users: it removes training-seen status as an explanation for discrimination, and it shows the fold-aligned ROC advantage is largely a *seen-versus-unseen-user* effect. (Unseen malicious users can still differ from unseen benign users in profile as well as behavior; the masking analysis in Section 6.7 probes whether the residual signal is behavioral.) The fold-aligned numbers below follow the local baselines' protocol for comparability, but should be read with this in mind. Scoring uses the full corpus, and fold masks are applied only afterward. We adopted this protocol after finding that an earlier pipeline, which re-filtered an already-filtered score file, silently dropped the benign validation slice and overstated the session LM's detector precision; all detector numbers reported here use the corrected full-population path.

5.2 Mechanistic estimands

Both estimands operate on *receivers* (positive user-day examples) and report effects of intervening on a selected sparse feature set S relative to a matched *active control* set C of equally many features drawn from those with nontrivial activation support (activation frequency at least 0.002). Inert controls make intervention effects look artificially large, so control activity is a precondition, not an afterthought (Zhang & Nanda, 2023).

Causal patching (sufficiency-style evidence). For each receiver we select donor user-days matched on an organizational context mode (role, department×role, project×role, team, or department), drawn either from benign users or—as a same-class control—from other anomalous user-days. At token positions where features in S are active, we patch the delta-SAE features toward the donor prototype and measure the resulting change in adapted NLL, keeping each receiver's best donor. Let Δ^S_{ben} and Δ^S_{anom} denote a receiver's best NLL change under benign and anomalous donors. The *repair advantage* of S is $\text{RA}(S) = \overline{\Delta^S_{\text{anom}}} - \overline{\Delta^S_{\text{ben}}}$, and the reported estimand is the paired contrast $\text{RA}(S) - \text{RA}(C)$ computed over receivers with complete top/control and benign/anomalous donor values. A positive value means the selected features carry anomaly-specific repairable structure beyond what matched control features carry.

Ablation (necessity-style evidence). For each positive receiver paired with a context-matched benign receiver, we ablate the selected features and measure the NLL change on both. The *necessity advantage* is the paired top-versus-control contrast of (benign minus positive) ablation effects: positive values mean ablating S disrupts anomaly evidence on positive examples more than on matched benign examples, and more than ablating C does. Throughout, patching establishes *repairability*—the selected features carry causally useful information given the rest of the computation—not literal sufficiency of the features in isolation, and ablation establishes dependence rather than strict necessity, since redundant pathways may exist.

Same-user exclusion. In both estimands, donors and benign matches from the receiver's own user are excluded. Without this restriction, patching can "repair" an anomalous day by borrowing the same user's benign signature, which inflates effects for reasons unrelated to the hypothesized mechanism. An audit of earlier permissive-donor runs found nontrivial same-user matching rates; all mechanistic results reported here come from reruns with same-user exclusion enforced and verified.

Uncertainty. Point estimates are complete paired contrasts, not differences of marginal means, which diverge when donor availability is asymmetric. Because multiple positive days from the same user are not independent, headline confidence intervals use a *user-level cluster bootstrap* (10,000 draws with the malicious user as the resampling unit); we additionally report per-user estimates and leave-one-user-out effects. Receiver-level (day-level) bootstrap intervals are narrower and are reported only as supplementary material. Context modes constitute a small multiplicity surface: we report all audited context modes rather than selecting a winner, quote per-context intervals without multiplicity adjustment, and treat cross-context consistency of sign—not any single best context—as the replication criterion; headline text names the best context only as an exemplar.

Table 1: Detector comparison on CERT under two protocols. Fold-aligned rows follow the baselines' protocol, in which the session LM (trained once on 90 percent of benign users) faces mostly training-seen benign test users while baselines exclude test users from training; bold marks the best fold-aligned value per metric (held-out rank: lower is better). The italicized user-disjoint rows restrict the LM's benign comparison population to never-trained validation users and are the fairer generalization test for the LM: its ranking advantage largely disappears (day ROC 0.954 to 0.545 on r6.2 and 0.959 to 0.668 on r4.2 computed on the full evaluation pool; the fold-aligned column uses the baselines' per-fold test populations, hence its slightly different 0.953/0.964), showing the fold-aligned strength is mostly a seen-versus-unseen-user effect; under the user-disjoint protocol the LM's day ROC falls at or below every baseline's fold-aligned value. User-disjoint PR values are computed at a very different prevalence (60 malicious versus 79 benign users on r4.2) and are not comparable to the fold-aligned column; the informative cross-protocol comparison is the ROC collapse. Baseline user-level ROC is not reported by the fold-aligned benchmark harness.

Dataset	Method	Day PR-AUC	Day ROC-AUC	User PR-AUC	User ROC-AUC	Held-out rank
r6.2	Session LM (adapted NLL)	0.0008	**0.953**	0.054	0.970	**24.0**
	Deep SVDD	**0.0115**	0.628	**0.211**	–	83.2
	GRU AE	0.0057	0.766	0.081	–	24.8
	LSTM AE	0.0021	0.768	0.057	–	24.2
	Isolation Forest	0.0002	0.713	0.013	–	153.0
	Session LM, user-disjoint benign	0.0005	0.545	0.013	0.546	–
r4.2	Session LM (adapted NLL)	0.0134	**0.964**	0.101	0.969	**26.4**
	Deep SVDD	**0.0337**	0.743	**0.382**	–	53.4
	GRU AE	0.0254	0.696	0.124	–	86.6
	LSTM AE	0.0236	0.714	0.120	–	92.1
	Isolation Forest	0.0003	0.715	0.008	–	186.4
	Session LM, user-disjoint benign	0.1023	0.668	0.521	0.565	–

6 Results

6.1 Detection: strong ranking, weak day-level precision

Table 1 reports the fold-aligned comparison. The adapted NLL score ranks strongly: day-level ROC-AUC is 0.953 on r6.2 and 0.964 on r4.2, user-level ROC-AUC is 0.97 on both, and the held-out malicious user ranks in the top 3.5% of 801 test users on average (mean rank 24.0 on r6.2, 26.4 on r4.2). Adaptation carries the signal: base-model NLL achieves day ROC-AUC of only 0.461 (r6.2) and 0.581 (r4.2). Day-level PR-AUC tells a different story: at extreme base rates ($\sim 10^{-5}$ positive on r6.2) the adapted model reaches only 0.00075 on r6.2 and 0.0134 on r4.2, below Deep SVDD (0.0115 and 0.0337) and the recurrent autoencoders, though above Isolation Forest on both datasets.

Figure 2 summarizes the pattern. Combined with the never-trained-benign sensitivity above, the score-level conclusion anticipates the mechanistic one: the model's ranking strength is largely user novelty rather than behavioral discrimination: on r6.2 almost entirely so, on r4.2 with a residual anomaly signal (0.668 day ROC against unseen benigns) whose behavioral character is supported by the feature attribution rather than by the ROC value itself. Stated plainly: under the user-disjoint comparison the session LM does *not* outperform the feature-based baselines: its day ROC (0.545/0.668) falls at or below every baseline's fold-aligned value (0.628–0.768), and the baselines already exclude test users from training. Within-user day ranking sharpens the picture. For each malicious user we computed the ROC-AUC of the adapted NLL over that user's own days (positive versus own-benign; identity held fixed by construction). The score does carry within-user signal on both benchmarks: mean within-user AUC 0.669 (95% user bootstrap $[0.619, 0.717]$; 52/60 users above chance) on r4.2 and 0.725 ($[0.654, 0.795]$; 4/4 users, e.g. the dominant user at 0.768) on r6.2. User novelty therefore explains the collapse of *cross-user* ranking—between-user profile novelty swamps the within-user signal when users are compared—rather than an absence of within-user behavioral signal; on r6.2 the latter is consistent with the secondary behavioral feature family recovered in Section 6.6. We therefore treat detection as background and ask what internal structure the benign-only adaptation induced.

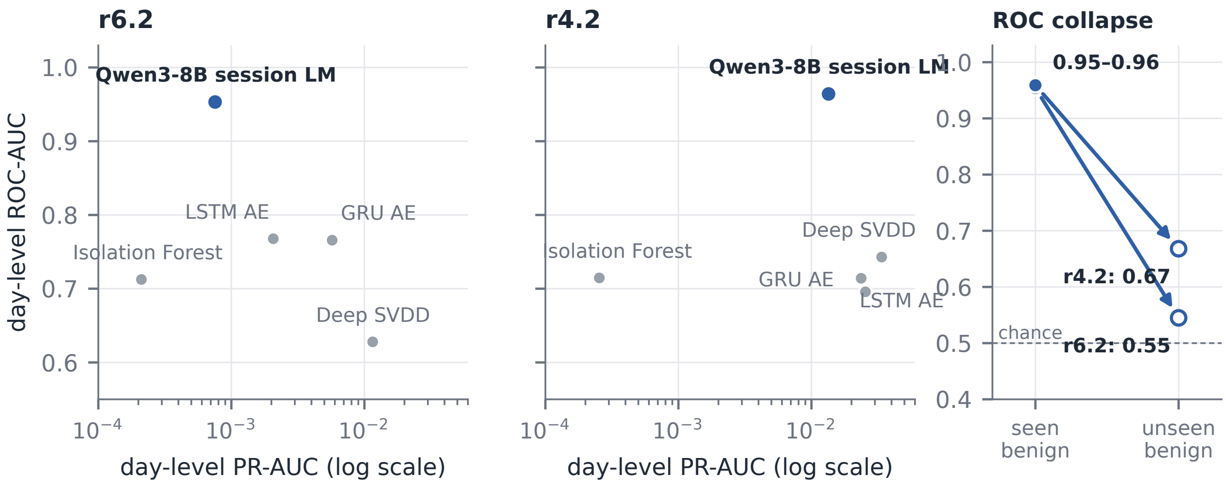

Figure 2: Score-level results. Left two panels: ranking–precision dissociation under the fold-aligned protocol; each point is one detector, and the adapted session LM (blue) attains the highest day-level ROC-AUC on both benchmarks while remaining mid-pack on day-level PR-AUC (log scale) at extreme base rates. Right panel: that apparent strength largely disappears when the benign comparison population is restricted to never-trained users (day ROC $0.954 \to 0.545$ on r6.2, $0.959 \to 0.668$ on r4.2; Section 6.1)—the paper's central score-level finding.

6.2 r6.2: necessity is consistent across users; causal effects concentrate in one user

Figure 3 shows all headline mechanistic effects. Because gap-ranked features carry more activation energy than active controls, the headline contrasts were re-verified against *dose-matched* controls (features matched on activation frequency, magnitude, and decoder norm; edit-size gap closed to 1.3–1.8×; Appendix A): all four contrasts remain positive, with the r4.2 causal contrast (+0.0017 $[0.0011, 0.0023]$) and the r6.2 necessity contrast (+0.061 $[0.007, 0.084]$) individually significant. r6.2 patching is the weakest link (positive but descriptive with four clusters); its causal evidence rests on necessity and the donor-type contrast. The 70 positive receiver-days on r6.2 come from only four malicious users, with highly unequal day counts (46/18/5/1), so we treat per-user structure as part of the result rather than pooling days as independent units. The necessity side is the strongest r6.2 finding: ablating the layer-18 top-5 features harms anomaly evidence more than ablating matched active controls for *every* malicious user in the project×role context (per-user effects 0.019–0.103; pooled 0.065; cluster interval $[0.026, 0.083]$, though with only four clusters such intervals are descriptive rather than conventionally inferential), and the role context is consistent as well (0.062, $[0.016, 0.083]$; three of four users positive). The causal patching side is positive but concentrated: the pooled role-context contrast is 0.0068 with cluster CI $[0.0001, 0.0100]$, positive for all four users individually, yet dominated by the most active user (0.010 versus 0.0001–0.0005 for the other three), and the department×role and project×role intervals cross zero under user-level resampling. The team context yields no finite causal estimate because same-user exclusion leaves no valid anomalous-donor comparison in that stratum; we report it as unavailable rather than imputing it.

Two comparisons calibrate these magnitudes. First, the strengthened protocol matters: same-user exclusion reduced the causal point estimates relative to earlier permissive-donor runs, so part of the earlier effect was inflated exactly as the audit hypothesized, but the effect does not vanish. Second, a matched session-autoencoder comparator evaluated on the same receivers at the same day-level unit yields a best contrast of 0.0011; because the comparator's score is on a different scale (autoencoder reconstruction error rather than NLL), we report this as context rather than as a ratio, and defer standardized cross-model effect sizes to future work.

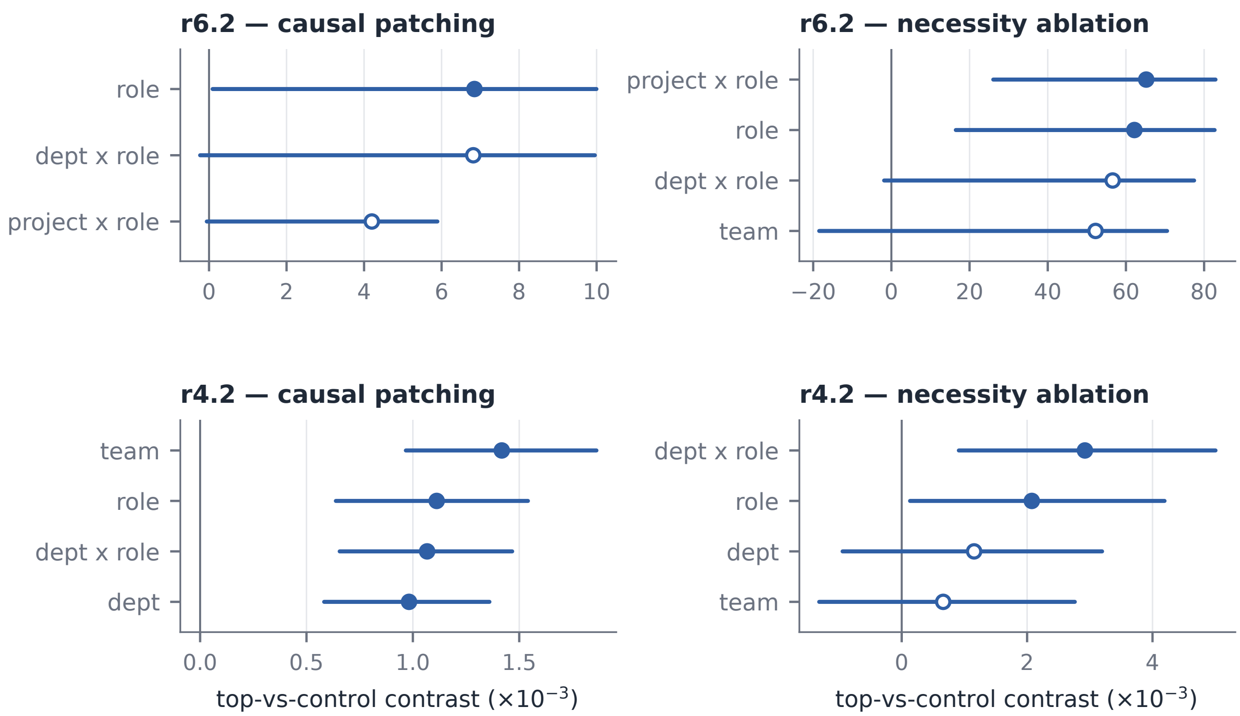

Figure 3: Mechanistic effects under same-user exclusion, by benchmark and estimand (note the panel-specific scales, all in units of 10^{-3} adapted-NLL). Points are paired complete-case top-versus-control contrasts; horizontal lines are 95% user-level cluster bootstrap intervals (10,000 draws with the malicious user as resampling unit; the prespecified primary analysis); open markers denote intervals that cross zero.

Because four users cannot support a split-half confirmation, we ran leave-one-user-out (LOUO) discovery/confirmation: for each fold, features are re-selected using only the other three users' positives and effects are estimated on the held-out user with the configuration frozen. Feature selection itself proved user-sensitive: the three folds that retain the dominant user select nearly identical top-5 sets, while the fold that holds out the dominant user selects a fully disjoint set: r6.2's feature identity is largely defined by that user's data. The causal effect nevertheless passes its hardest test: the disjoint features, chosen from the other users' 24 days, produce a significant repair effect on the dominant user's 46 held-out days (+0.0037 to +0.0038 across contexts, e.g. project×role $[0.0020, 0.0056]$), while held-out effects on the low-activity users are near zero. Held-out necessity is asymmetric: it replicates on the two multi-day low-activity users (up to +0.043, $[0.030, 0.055]$) but is negative on the dominant-user fold. Causal repairability therefore transfers to the dominant held-out user while ablation dependence is user-specific: the SAE space contains multiple causally effective repair directions discoverable from disjoint user subsets, but which features a given user's anomaly evidence depends on varies by user.

6.3 r4.2: direct transfer fails; native rediscovery succeeds

Applying the r6.2 mechanism configuration (layer 18, its SAE, and its feature sets) directly to r4.2 produces negative top-versus-control contrasts across all audited transfer configurations under full uncapped evaluation on all 1309 receivers: the best transferred configuration per sparsity setting reaches -0.00094 $[-0.00117, -0.00072]$, -0.00105 $[-0.00149, -0.00064]$, and -0.00113 $[-0.00144, -0.00082]$. Direct transfer of the r6.2 sparse feature set to r4.2 thus fails under every audited configuration: at the level of literal feature identity, nothing transfers in this training run.

An exploratory search over layers and SAE hyperparameters on r4.2, however, identifies a candidate mechanism at layer 26 with a smaller dictionary ($m = 2$, $k = 4$). Under same-user exclusion, on populations

Table 2: Mechanistic summary (best context mode per estimand). All rows use the same-user-excluded protocol with active-control feature sets; the transferred row applies the r6.2 layer-18 configuration to r4.2 without re-fitting. Effects are paired complete-case top-versus-control contrasts; intervals are the prespecified user-level cluster bootstrap (10,000 draws, malicious user as resampling unit; descriptive for r6.2's four clusters).

Dataset	Estimand	Context	Effect	95% CI	Note
r6.2	Token-SAE causal	role	0.006848	[0.000092, 0.009996]	4/4 users positive
r6.2	Token-SAE necessity	project×role	0.065188	[0.026059, 0.082920]	4/4 users positive
r4.2	Transferred causal	multiple	< 0	all audited configs < 0	direct transfer fails
r4.2	Native token-SAE causal	team	0.001418	[0.000967, 0.001863]	all contexts positive
r4.2	Native token-SAE necessity	dept×role	0.002922	[0.000911, 0.005005]	necessity partial

of 1039–1301 available receivers from 49–54 malicious users per context mode, causal contrasts are positive with user-level cluster confidence intervals excluding zero for all four context modes: team 0.00142 $[0.00097, 0.00186]$, role 0.00111 $[0.00064, 0.00154]$, department×role 0.00107 $[0.00066, 0.00147]$, and department 0.00098 $[0.00058, 0.00136]$; mean-of-user-means intervals also exclude zero. The session-autoencoder comparator's best day-level contrast on the same receivers is 0.00091 (different score scale; context only). Necessity is positive but partial: department×role (0.0029, $[0.0009, 0.0050]$) and role (0.0021, $[0.0001, 0.0042]$) remain positive under clustering, while department and team cross zero.

Because this configuration was selected by searching the same positive population it is evaluated on, we ran a held-out confirmation. The malicious users were split 30/30 (by seeded shuffle; 671 versus 638 positive days); features were re-selected using discovery-user positives only—the resulting top-5 set shares four of five features with the full-population selection, indicating stable selection across disjoint user subsets—and effects were then estimated exclusively on the 30 held-out users with the configuration frozen. Under the prespecified primary analysis (user-level cluster bootstrap), all four causal point estimates are positive on held-out users at roughly half the exploratory magnitude (0.00034–0.00076)—the expected signature of selection inflation—but only the department context individually excludes zero (0.00076, cluster CI $[0.00035, 0.00121]$; 21 of 29 users positive). We read this as *directional replication* with one individually significant context, not as confirmation of all four. Held-out necessity is directionally positive for department×role and role, while team is a confirmed null. Two caveats bound the claim: the SAE configuration (layer, dictionary size, sparsity) was fixed by the earlier full-population search, so independence holds at the feature level only; and secondary uncertainty procedures (receiver-level, mean-of-user-means) give more favorable readings that we report in supplementary material rather than in the headline.

6.4 What the mechanisms encode: profile novelty versus behavior

The estimands above establish that causally relevant sparse structure exists; they do not say what it represents. We therefore attribute each headline feature's post-top-k activation mass to the serialization's line classes (`DAY` organizational header, `PSY` psychometrics, `SES` behavioral session events) on all positive examples, with a benign contrast set, normalizing by each class's token share (Table 3).

The result is unambiguous and opposite between benchmarks. The benign contrast sharpens it: on benign examples the same r6.2 features place essentially zero mass on `PSY` tokens (e.g., the top feature: 0% benign versus 99.8% positive), so profile-token firing is specific to unfamiliar profiles rather than to psychometric tokens per se. On r6.2, the five top causal features place 99.8–100% of their activation mass on profile tokens—four on the `PSY` line (enrichment $\approx 13\times$ over token share; the highest-activating tokens are the psychometric digit values themselves) and one on the `DAY` header—while behavioral `SES` tokens, 78% of the input, carry $\leq 0.2\%$ of the mass. The r6.2 mechanism is *profile-bound*, in a pattern consistent with user-profile novelty: the benign-only adapter learned the benign population's profile distribution, and unseen malicious users' profile fields produce large adapter deltas. This single fact retroactively explains the r6.2 phenomenology: causal repair concentrates in the dominant user (patching profile tokens toward a context-

Table 3: Token attribution of the top-5 causal features (positive examples). Columns are activation-mass fractions by serialization line class; PSY enrich is mass fraction over token share for the psychometric line. r6.2 features are profile-bound; four of five r4.2 features are behavioral (SES enrichment 1.33x over a 0.75 token share).

Bench	Feature	PSY mass	DAY mass	SES mass	PSY enrich	Top tokens
r6.2	14358	0.998	0.000	0.002	13.4×	psychometric values
r6.2	12848	0.999	0.000	0.001	13.4×	psychometric values
r6.2	4196	1.000	0.000	0.000	13.4×	psychometric values
r6.2	13580	0.000	0.999	0.001	≈0	org header
r6.2	11292	1.000	0.000	0.000	13.4×	psychometric values
r4.2	4596	0.001	0.000	0.999	≈0	session values
r4.2	3673	0.000	0.000	1.000	≈0	session durations
r4.2	2302	0.001	0.998	0.001	≈0	org header
r4.2	3455	0.000	0.000	1.000	≈0	session values
r4.2	1268	0.000	0.000	1.000	≈0	session values

matched donor makes the profile look familiar), necessity does not transfer across users (each user's profile is constant), and feature identity flips when the dominant user leaves the selection set.

On r4.2, four of the five top features are *behavioral*: they place 99.9–100% of their mass on `SES` tokens (enrichment 1.33× over the 75% token share) with essentially zero psychometric mass, and one fires specifically on session-duration fields; the fifth is `DAY`-bound. A natural first hypothesis for the contrast is the malicious population's structure—with four positive users, identity is the cheapest separator; with sixty, behavioral regularities carry the evidence—but Section 6.6 tests this directly and rejects its strong form, pointing instead to finer benchmark properties.

6.5 Seed-stability controls: the dissociation is not dictionary noise

Sparse dictionaries are not identifiable: literal feature coordinates can fail to transfer even between two SAE seeds on the same data, so cross-benchmark transfer failure could in principle say nothing about benchmarks. We retrained the headline SAE configuration with two fresh seeds per benchmark on the same cached deltas. Because every SAE decodes into the same d_{model} residual space, feature correspondence is measurable as decoder cosine similarity (Table 4): within a benchmark, each seed's top-5 features have close counterparts in every other seed's dictionary ($|\cos|$ 0.81–0.95 against empirical dictionary-wide best-match dictionary-median null baselines of 0.72–0.82); across benchmarks, top-5 best matches sit at or below their nulls (0.22–0.24 against nulls of 0.22–0.40 at native layers) in both directions. All geometry is computed in original residual-stream coordinates: decoder columns are mapped back through each SAE's per-coordinate standardization scale before normalization, since raw standardized columns from dictionaries with different scales are not mutually comparable (the raw-coordinate comparison, our earlier convention, gives the same qualitative pattern). Causal rediscovery is likewise seed-robust: fresh-seed SAEs with their own re-selected features reproduce positive r6.2 contrasts in every finite context (e.g., role 0.0071 and 0.0019 under seeds 43 and 44, versus 0.0068 originally). The profile-bound and behavioral feature families are each stable across SAE initializations for the same cached adapter representations (adapter-seed robustness remains untested); their non-correspondence across benchmarks is a property of the benchmark–adapter pairs. The native-layer cross-benchmark comparison (18 versus 26) sits at or below the empirical null, but that partly reflects a depth difference. The matched-layer comparison at layer 18 is sharper: overall dictionary geometry is substantially shared there (whole-dictionary median best-match up to 0.96, as expected since both SAEs encode the same base model's layer-18 residual stream), yet the *top-5* features align at or below those medians (0.30 and 0.65): the anomaly-relevant directions are more benchmark-specific than typical dictionary directions. Matched-layer top-5 alignment also remains below within-benchmark cross-seed alignment (0.81–0.95), so the distinct headline mechanisms are not consistent with a simple feature relabeling of one shared sparse representation, even after controlling for depth. (r4.2's headline mechanism is itself at layer 26; its layer-18 features used here are off-mechanism, which further limits the correspondence one should expect.)

Table 4: Feature alignment in residual-stream coordinates (decoder columns mapped back through each SAE's per-coordinate standardization before normalization). Best-match $|\cos|$ of each source SAE's top-5 features into a target dictionary, versus the whole-dictionary median (empirical null). Within-benchmark cross-seed alignment exceeds the null; cross-benchmark top-5 alignment sits at or below its null—at matched layer the top features align worse than typical dictionary features despite substantially shared overall geometry.

Comparison	Top-5 best-match $\lvert\cos\rvert$	Empirical null baseline
within r6.2 (3 seed pairs)	0.807–0.856	0.720–0.750
within r4.2 (3 seed pairs)	0.837–0.946	0.814–0.818
across benchmarks, native layers (18 vs 26)	0.218–0.236	0.224–0.405
across benchmarks, matched layer 18	0.295–0.652	0.370–0.963

6.6 Dictionary and population robustness

The picture so far (Table 6): benign-only adaptation induces sparse token-level structure that is causally implicated in anomaly evidence on both benchmarks, with held-out replication that is directional on r4.2 and dominant-user-concentrated on r6.2; the structure is benchmark-specific not merely in coordinates but in *kind*: identity metadata on r6.2, behavior on r4.2. This subsection stress-tests that picture against the two remaining artifact explanations: the positive population and the dictionary. We tested the population-size hypothesis directly by malicious-population subsampling on r4.2: holding the model, benchmark, SAE dictionary, and benign population fixed, we resampled positive populations of $K = 4, 10, 20, 30, 60$ users (ten repetitions each) and re-ranked features for each. The strong form of the hypothesis is *rejected at the feature-selection stage*: profile-mass share among top-5 features is nearly flat in K (0.25 at K=4 versus 0.16–0.20 at $K \geq 10$; typically one profile-bound feature in five at every size), nowhere near r6.2's near-total profile dominance. Small malicious populations tilt selection toward profile features only mildly. Because the shared dictionary was itself fitted on the full positive population, this experiment isolates the selection stage conditional on the fixed r4.2 dictionary; it does not test whether positive-population size could influence dictionary formation itself. A stronger check removes positives from dictionary formation altogether: we retrained both SAEs from scratch on benign token deltas only and reran the full downstream pipeline: feature reselection under held-out discipline (discovery users on r4.2; LOUO on r6.2), token attribution, and confirmation evaluation. The dissociation reproduces at the dictionary level. On r4.2, the benign-dictionary top-5 places $\geq 94\%$ of activation mass on behavioral `SES` tokens (four of five features at 100%, profile mass ≈ 0); on r6.2, every LOUO fold's top-5 is $\geq 99\%$ profile-bound, three of four folds select an identical feature set, and the fold excluding the dominant user diverges wholesale, mirroring the original dictionary's dominant-user-defined feature identity (Table 5). Held-out intervention effects are estimand-dependent under the new dictionaries: r4.2 necessity replicates across all four contexts (+0.004 to +0.013; 528–638 complete pairs), while its patch-repair contrast does not (−0.0002 to −0.0004); on r6.2, causal repair is positive on one held-out fold and necessity on the dominant-user fold. What the selected features *encode* is therefore dictionary-independent; which intervention estimand replicates on which held-out users is not (Table 5). r6.2's profile capture therefore reflects benchmark-specific factors—plausibly the distinctiveness of its four malicious users' profile fields within its benign profile distribution, and properties of its adapter's learned geometry—and identifying which benchmark properties drive metadata capture is an open question the audit makes precise.

Finally, a smaller-scale rerun tests scale-, serializer-, and adapter-seed-dependence jointly. We repaired the serializer defect (restoring the four omitted file-routing fields on r6.2; r4.2's release contains one of the four), retrained Qwen2.5-3B QLoRA adapters from scratch with two adapter seeds per benchmark (one epoch; full 287,827-day corpus on r4.2, 300k of 1.25M days on r6.2), extracted fresh token deltas at layers 12/18/24, retrained benign-only dictionaries, and reran selection and attribution. The dissociation reproduces under both adapter seeds. On r4.2, the selected features are behavioral in every audited configuration (five of five features ≥ 0.94 SES mass at the best layer under both seeds), and the adapters retain unseen-user discrimination (day ROC 0.678/0.718 against never-trained benigns, versus 0.668 at 8B). On r6.2, profile-bound features dominate at the deepest extracted layer under both seeds (three of five and four of five

Table 5: Dictionary-independence summary. Attribution (what the selected features encode) reproduces exactly under SAE dictionaries retrained on benign rows only, while held-out replication of the intervention estimands is estimand- and dictionary-dependent.

Dataset	Dictionary	Attribution	Held-out patching	Held-out ablation
r6.2	Full-pool	Profile-bound (99.8–100%)	Dominant-user concentrated	User-specific (negative on dominant fold)
r6.2	Benign-only	≥99% profile-bound, all folds	Positive on one held-out fold	Positive on dominant-user fold
r4.2	Full-pool	4/5 behavioral	Directional; one context significant	Partial
r4.2	Benign-only	≥94% behavioral	Does not replicate	Replicates in all four contexts

features at $\approx$ 100% profile mass), while shallower layers lean behavioral: at this scale, profile capture concentrates at depth. The two r6.2 adapters differ in quality (day ROC 0.414 versus 0.738), and profile dominance is strongest in the better-trained seed, consistent with profile capture being acquired through adaptation rather than induced by the serialization defect or particular to the 8B run. Rerunning the headline intervention estimands on all four 3B adapters replicates the estimand asymmetry observed under the 8B benign dictionaries: r4.2 necessity-style dependence replicates across both adapter seeds with nearly identical effects (+0.00218 and +0.00215; 1301 complete pairs each) while its patch-repair contrast does not (−0.0004 under both seeds), and r6.2's intervention effects remain seed- and estimand-idiosyncratic (causal +0.0175 under one seed, near zero under the other). Ablation dependence is thus the seed-robust causal signature of the behavioral family.

Excluding profile-dominated features from selection likewise reveals that r6.2 also contains a *behavioral* feature family: SES-restricted ranking under LOUO discipline selects stable behavioral features (two of the top five recur across three of four folds, all with near-zero profile mass), and these features carry held-out causal effect on the dominant user (+0.0027 to +0.0029 across contexts, roughly three-quarters of the profile-feature effect), with partial held-out necessity (positive on the small multi-day users; role-context positive on the dominant-user fold, where the profile features' necessity was uniformly negative). r6.2's profile capture is therefore a property of unconstrained feature *selection*, not an absence of behavioral causal structure, which is precisely the kind of distinction the audit exists to make.

6.7 Score-level profile counterfactuals

A score-time ablation asks what the *score* is sensitive to. We re-serialized the evaluation population without the `PSY` line (*no-psy*) and additionally with organizational identifiers masked (*no-profile*), and rescored with the frozen models on a fixed unseen-versus-unseen rescoring population: all positive user-days plus every never-trained benign user (on r4.2: 89 benign users—the 79 validation-split users plus 10 never-trained others—for 42,468 day rows from 149 users), which is why the original-arm values here differ slightly from the val-only analysis of Section 6.1 (day ROC 0.670 versus 0.668; user ROC 0.666 versus 0.565, the latter sensitive to the ten added users under max-aggregation). Uncertainty for the masking contrasts uses the same user-level cluster bootstrap as the mechanistic estimands (10,000 draws, users as resampling units stratified by label, paired within draws). On r6.2, all three variants remain near chance at the day level (ROC 0.547 original, 0.575 no-psy, 0.471 no-profile) and no masking change is significant (all ΔROC intervals span zero; descriptive with four malicious clusters): with every user unseen, profile novelty separates nothing, and no clear behavioral signal is revealed underneath. On r4.2, masking psychometrics *improves* discrimination substantially (day ROC $0.670 \rightarrow 0.797$, ΔROC +0.127 $[0.061, 0.193]$; day PR $0.069 \rightarrow 0.112$; user ROC $0.666 \rightarrow 0.775$, ΔROC +0.108 $[0.038, 0.179]$; masking all organizational identifiers as well yields a smaller, nonsignificant day-level gain of +0.063 $[-0.003, 0.130]$): the simulated psychometric fields were not merely uninformative but actively obscured behavioral signal. Static profile metadata in the serialization is thus a liability twice over: it invites identity capture during selection and degrades behavior-based scoring even when the mechanism is behavioral.

A swap-counterfactual experiment sharpens what the score itself is sensitive to. For every positive user-day we constructed two variants—profile fields replaced by a random benign (validation-split) user's, and conversely the entire behavioral content replaced while keeping the profile—and rescored with the frozen models. On *both* benchmarks, the score is more disturbed by profile perturbation than by replacing all behavior (mean ΔNLL +0.155 versus +0.072 on r6.2, ratio 2.2; +0.180 versus +0.102 on r4.2, ratio 1.8). Because donors are unseen users, this measures perturbation sensitivity rather than profile familiarity, and it shows that profile tokens dominate the raw NLL surface of both adapters, consistent with masking *improving* r4.2 discrimination by removing profile-driven score variance. The benchmarks thus differ not in raw score sensitivity, which is profile-dominated everywhere, but in where the anomaly-*discriminative* signal lives: a distinction that feature-level attribution and masking expose and that score-level perturbation alone cannot.

6.8 Real-data replication with real psychometrics (TWOS)

All results above are on simulated CERT data, and the population-structure account makes a falsifiable prediction about real data: when malicious users are *not* a small distinctive minority, identity is a poor separator and selection should be behavioral. TWOS (Harilal et al., 2017) provides exactly this test: 24 real users over five days, staged masquerade sessions on 12 of them (50% prevalence) plus five staged insider ("traitor") sessions, and—critically—each user's *real* IPIP-50 Big-Five inventory, giving a genuine psychometric `PSY` line where CERT's is simulated. We serialized mouse, keystroke, and login telemetry into CERT-analogous ten-minute windows (18,048 windows; 138 positive, of which 101 are masquerade and 37 traitor windows; 12-user benign train pool), ran the identical pipeline at smaller scale (Qwen2.5-3B benign-only QLoRA, two adapter seeds; benign-only dictionaries; full-pool selection; attribution), and reran the headline intervention estimands.

The audit certifies a clean detector on this benchmark: top-5 attribution is essentially 100% behavioral under both adapter seeds at both audited layers (worst single feature: 0.983 SES mass), despite real personality scores being present in every window; the score carries real signal (mean within-user ROC 0.783 across the 16 attacked users—12 masqueraded plus 4 traitors with logged activity—identity held fixed by construction); and the intervention estimands reproduce the estimand asymmetry a fourth time (ablation dependence +0.0019, patch-repair −0.0018; 138 receivers, team-matched donors, same-user excluded). This is the predicted outcome for the high-prevalence corner, and it complements the r6.2 corner (four highly distinctive users among thousands) where profile capture wins: the dissociation's behavioral pole is not an artifact of synthetic data or simulated psychometrics. Deeper analyses of this benchmark (insider-type boundaries, detector baselines, per-user failure cases) are developed in companion work.

7 Discussion

Audits, not leaderboards. Every component of this audit was necessary to reach the finding. Score-level evaluation showed strong ROC on both benchmarks; the intervention estimands established causal relevance under at least one audited estimand on both; the confirmation splits showed directional held-out replication; the seed controls showed stability across SAE initializations. Only token-level attribution revealed that one of them scores evidence consistent with *who a user is* rather than *what they did*. Any prefix of the audit would have licensed a stronger—and wrong—behavioral interpretation. We take this as the central methodological lesson: for anomaly models trained on metadata-rich telemetry, causal validation and semantic attribution are separate obligations, and only their conjunction says what a detector detects. In this sense the audit's mechanistic layer is demonstrably necessary—every cheaper prefix misleads, and on r6.2 the black-box probes were silent where attribution was decisive—while never sufficient alone: only conjoined with causal validation and score-level evaluation does it identify what a detector detects. One might object that we audit a detector of our own construction; but benign-only adaptation is the natural one-class recipe for the LoRA-tuned insider-detection systems now appearing (Kong et al., 2025), and the failure modes exposed here are properties of metadata-rich serialization and of evaluation protocol—both inherited from existing practice—rather than of any particular adapter architecture. A cross-architecture probe sharpens this: classical one-class detectors (Isolation Forest, one-class SVM, PCA reconstruction) trained on the same r6.2 day-level features *with profile columns included* place only 4–7% of permutation-importance mass on

Table 6: Audit claim map: what the evidence supports, rejects, and fails to support.

Claim	Status
Benign-only QLoRA training is valid one-class training	Supported
Fold-aligned detector strength reflects behavioral discrimination	Rejected (seen-vs-unseen-user effect)
r6.2 contains a profile-bound sparse feature family with causal relevance under the audited estimands	Supported descriptively; held-out replication concentrates on the dominant user; necessity significant under dose-matched controls (App. A.5)
r4.2 contains a behavior-associated sparse feature family supported by patching and ablation	Supported; causal contrast significant under dose-matched controls; ablation replication persists under a benign-only dictionary and across 3B adapter seeds, patch-repair replication does not
Configuration-independent r4.2 confirmation	Not established
Literal feature transfer across benchmarks succeeds	Rejected
Transfer failure is explained by SAE seed non-identifiability	Rejected (alignment controls)
Positive-population size alone explains the feature-selection dissociation (conditional on the fixed r4.2 dictionary)	Rejected at the selection stage (subsampling)
The profile/behavior attribution dissociation requires positive-shaped dictionaries	Rejected: benign-only SAE retrain reproduces it (r4.2 top-5 $\geq$94% behavioral; r6.2 $\geq$99% profile-bound in all LOUO folds)
The dissociation is particular to the 8B run, the serializer defect, or one adapter seed	Rejected: 3B repaired-serializer rerun reproduces attribution across two adapter seeds per benchmark, and r4.2 ablation dependence replicates across 3B seeds
The behavioral pole is an artifact of synthetic data or simulated psychometrics	Rejected: TWOS (real users, real Big-Five) yields 100% behavioral attribution across two adapter seeds at 50% malicious prevalence, as the population account predicts
Profile capture is a generic dataset property that any detector class inherits	Rejected at attribution level: classical one-class detectors on the same r6.2 features place 4–7% importance on profile, select no profile feature in any top-5, and lose nothing when profile is removed

profile fields, select no profile feature in any top-5, and lose nothing when profile columns are removed. The profile shortcut is therefore not a generic dataset property inherited by any detector: it emerges in the sequence-likelihood pathway, where per-user-constant profile tokens are exactly what a next-token objective is incentivized to compress. The detector class the field is now adopting carries a shortcut risk its classical predecessors demonstrably did not—which is why auditing it matters. The refined account also makes a testable prediction we leave to future work: sequence autoencoders, which share the sequence-prediction objective but not the pretrained representation, should fall between the tabular detectors and the LM in profile reliance; auditing their latent spaces with this framework would locate the shortcut's origin more finely. Auditing the class before it is deployed is the point.

A failure mode without a simple boundary condition. The profile-novelty mechanism is not a bug in the model; it is a solution to r6.2's separation problem. But the obvious explanation—population size—fails our direct test at the feature-selection stage: four-user subsamples of r4.2 still select behavioral features. Whatever drives metadata capture on r6.2 involves finer benchmark properties (how distinctive the malicious users' profile fields are within the benign profile distribution; what the benign-only adapter happened to compress into its deltas)—and, per the cross-architecture probe, properties of the sequence-likelihood pathway specifically, since classical detectors on the same features do not take the shortcut. Neither factor of the two-factor account has yet been manipulated in isolation: the classical detectors differ from the language model in objective, representation, architecture, and capacity simultaneously, and profile distinctiveness has not been varied independently of malicious-user count; factorial designs (matched adapters trained with profile fields removed, shuffled, or loss-downweighted; distinctiveness varied at fixed count) remain the decisive tests of the account. The account's two corners are now both instantiated: profile capture where a tiny distinctive malicious minority makes identity cheap (r6.2), and cleanly behavioral selection where half the population is malicious and identity separates nothing (TWOS, on real users with real psychometrics; Section 6.8). The cross-architecture probe adds the second factor: classical one-class detectors on the same r6.2 features do not take the shortcut, so "cheap" is relative to the learner. The account in its refined form is two-factor—*population structure determines when the shortcut is available; the objective and representation determine which detectors take it.* Twelve profile columns among 242 are not especially cheap for a density model, but per-user-constant profile tokens are maximally cheap for a sequence-likelihood objective. This also explains a number reported earlier: under user-disjoint evaluation the session LM *underperforms* the classical baselines on r6.2 (0.545 against 0.63–0.77), exactly as expected if adaptation spent capacity on a shortcut that does not generalize while the baselines learned behavior. For benchmark design the implication is accordingly conditional but sharp: exposing static per-user metadata, especially simulated psychometrics with no operational counterpart, creates a capture risk for sequence-likelihood detectors that cannot be predicted from population size and that score-level evaluation will not notice—it must be audited for. For deployment, the implication is unchanged and sharp: an identity-proxy detector generalizes to *new users being unusual*, not to insider behavior.

Benchmark-specificity, revisited. Cross-benchmark transfer of literal features fails while native rediscovery succeeds on both sides, and the seed controls locate the failure in the benchmarks rather than in dictionary randomness. The universality question familiar from interpretability (Olah et al., 2020; Chughtai et al., 2023) thus reappears here with a mechanistic finding: we find no evidence that the two benchmark-specific adapters implement their anomaly signals through directly corresponding sparse feature families; indeed, the selected evidence differs in *kind*. What transfers is the audit itself—the pipeline that finds, validates, and identifies mechanisms—not the mechanisms.

Why the strengthened protocol is load-bearing. Four protocol elements changed conclusions materially during this work: full-population scoring (an earlier re-filtering bug overstated detector precision), same-user donor exclusion (permissive matching inflated causal effects), user-level cluster uncertainty and discovery/confirmation splits (receiver-level intervals and search-then-evaluate selection overstated significance), and token-level attribution (without it, the profile mechanism reads as behavioral). None are exotic; all are easy to omit silently. We report them as part of the contribution because intervention-based interpretability applied to security data inherits both the pitfalls of interpretability (Heimersheim & Nanda, 2024) and the base-rate pathologies of anomaly detection (Chandola et al., 2009).

8 Limitations

Our evaluation population is the matched active-session domain, not the full CERT answer key; incidents outside session coverage are invisible to all compared methods. The CERT benchmarks are simulations, and the psychometric and organizational fields they expose have no real-world measurement counterpart; results may not transfer to operational telemetry. The serializer omits four file-*routing* sub-count fields (to-USB, from-USB, file-activity-type, and disk transfer counts) due to a column-naming constraint, while retaining the principal behavioral signals: total USB events and mean USB session duration, total and executable file counts, e-mail volumes and attachment statistics, off-hours and weekend flags, session durations, and HTTP category counts including leak- and hacking-site visits. Claims about the mechanism concern this serialized field set; the repaired-serializer rerun (Section 6.6) shows the attribution dissociation does not depend on the omission. Day-level PR-AUC of the adapted session LM is weak in absolute terms at realistic base rates.

r6.2 mechanistic estimates rest on 70 receivers from 4 malicious users with highly unequal day counts, so its causal effect is dominated by a single user and one context mode (team) is unavailable under strict donor rules; we report cluster intervals, per-user effects, and leave-one-user-out results for this reason. The r4.2 native configuration was found by search, so we report a dictionary-conditional held-out-user confirmation alongside the exploratory estimates; the confirmed effect sizes are roughly half the exploratory ones, and only the department context is individually significant under the strictest user-level resampling. For r6.2, the analogous leave-one-user-out confirmation recovers a significant held-out causal effect on the dominant user but near-zero causal effects on the low-activity users, and held-out necessity does not transfer to the dominant user; r6.2 claims should be read at that resolution. All mechanistic results concern feature-level interventions at a single layer under a specific SAE family; they localize causal structure but do not constitute a complete circuit-level account (Chan et al., 2022). Attribution is based on activation mass over line classes; it identifies where features fire rather than fully specifying their computation, and the masking and swap experiments of Section 6.7 provide score-level profile-sensitivity evidence without identifying individual feature semantics. The SAE dictionaries were fitted on the full evaluation pool (including positive rows), so most held-out analyses are dictionary-conditional; the benign-only retraining check (Section 6.6) establishes that the attribution dissociation and r4.2's necessity-style replication are dictionary-independent, while patch-repair replication on held-out users is dictionary-sensitive and should be read as configuration-specific. SAE-seed robustness is established (Section 6.5); the smaller-scale rerun adds attribution-level replication across two adapter seeds per benchmark and a repaired serializer, and the headline intervention estimands rerun on the four 3B adapters replicate r4.2's ablation dependence across seeds; still, the 8B intervention estimands rest on one adaptation run per benchmark and the rerun's r6.2 arms train on a capped corpus, so the 8B causal claims remain scoped to their runs. Relatedly, the discovery/confirmation splits reselect feature identities but inherit configuration choices (layer, dictionary size, sparsity, context menu, intervention grid) made on the full population; confirmation is feature-level, not configuration-level. Finally, the population-structure explanation for the profile/behavior dissociation is supported by two synthetic benchmarks plus the TWOS replication (Section 6.8), which supplies the high-prevalence, real-psychometrics corner; what remains untested is the profile-capture corner on real data—an organization-scale population with a small, profile-distinctive malicious minority—which no public real dataset currently provides.

9 Broader Impact Statement

Insider-threat detection can reduce harm from credential misuse and data exfiltration, but it is also employee surveillance, and interpretability tools cut both ways: mechanisms that make alerts auditable can also make monitoring more invasive or lend unearned credibility to weak evidence. Two properties of this work partially mitigate these risks: training is benign-only, so the system models normal behavior rather than profiling individuals against incident templates, and the mechanistic protocol is designed to make claims falsifiable (active controls, exclusion rules, prespecified primary estimands) rather than persuasive. Nothing here justifies autonomous action against individuals. Deployments would require human review, documented appeal processes, and governance appropriate to workplace monitoring, and our results are on simulated data: they establish methodology, not operational readiness.

Reproducibility Statement

All experiments derive from tracked artifacts in the project repository: dataset construction and split assignment (`build_session_jsonl.py`, deterministic user-hash splits), training configurations (Section 4 lists all hyperparameters), full-population scoring and fold-aligned detector evaluation (fixed fold seed, 800 benign test users per fold), and causal/necessity evaluation with bootstrap reports whose estimands match the reported tables. Dataset rebuilds were verified byte-identical from checksummed transfer shards, and final mechanistic artifacts were audited for zero same-user donor rows and non-inert controls. An anonymized code and artifact package accompanies the submission as supplementary material and will be released publicly upon publication.

References

Daniel Arp, Erwin Quiring, Feargus Pendlebury, Alexander Warnecke, Fabio Pierazzi, Christian Wressnegger, Lorenzo Cavallaro, and Konrad Rieck. Dos and don'ts of machine learning in computer security. In *31st USENIX Security Symposium (USENIX Security 22)*, pp. 3971–3988, 2022.

Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nicholas L. Turner, et al. Towards monosemanticity: Decomposing language models with dictionary learning, 2023. Transformer Circuits Thread. `https://transformer-circuits.pub/2023/monosemantic-features`.

Lawrence Chan, Adrià Garriga-Alonso, Nicholas Goldowsky-Dill, Ryan Greenblatt, Jenny Nitishinskaya, Ansh Radhakrishnan, Buck Shlegeris, and Nate Thomas. Causal scrubbing: A method for rigorously testing interpretability hypotheses, 2022. Alignment Forum.

Varun Chandola, Arindam Banerjee, and Vipin Kumar. Anomaly detection: A survey. *ACM Computing Surveys*, 41(3):15:1–15:58, 2009.

David Chanin, James Wilken-Smith, Tomáš Dulka, Hardik Bhatnagar, and Joseph Bloom. A is for absorption: Studying feature splitting and absorption in sparse autoencoders. *arXiv preprint arXiv:2409.14507*, 2024.

Bilal Chughtai, Lawrence Chan, and Neel Nanda. A toy model of universality: Reverse engineering how networks learn group operations. In *International Conference on Machine Learning*, 2023.

Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models, 2023. arXiv preprint arXiv:2309.08600.

Alexander D'Amour, Katherine Heller, Dan Moldovan, et al. Underspecification presents challenges for credibility in modern machine learning. *Journal of Machine Learning Research*, 23(226):1–61, 2022.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. Qlora: Efficient finetuning of quantized llms. In *Advances in Neural Information Processing Systems*, 2023.

Min Du, Feifei Li, Guineng Zheng, and Vivek Srikumar. Deeplog: Anomaly detection and diagnosis from system logs through deep learning. In *ACM SIGSAC Conference on Computer and Communications Security*, 2017.

Leo Gao, Tom Dupré de la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders, 2024. arXiv preprint arXiv:2406.04093.

Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673, 2020.

Joshua Glasser and Brian Lindauer. Bridging the gap: A pragmatic approach to generating insider threat data. In *IEEE Security and Privacy Workshops*, 2013.

Haixuan Guo, Shuhan Yuan, and Xintao Wu. Logbert: Log anomaly detection via bert. In *International Joint Conference on Neural Networks*, 2021.

Athul Harilal, Flavio Toffalini, John Castellanos, Juan Guarnizo, Ivan Homoliak, and Martín Ochoa. Twos: A dataset of malicious insider threat behavior based on a gamified competition. In *Proceedings of the 2017 International Workshop on Managing Insider Security Threats*, pp. 45–56, 2017.

Stefan Heimersheim and Neel Nanda. How to use and interpret activation patching, 2024. arXiv preprint arXiv:2404.15255.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.

Jacob Kauffmann, Lukas Ruff, Grégoire Montavon, and Klaus-Robert Müller. The clever hans effect in anomaly detection. *arXiv preprint arXiv:2006.10609*, 2020.

Kaichuan Kong, Dongjie Liu, Xiaobo Jin, Guanggang Geng, Zhiying Li, and Jian Weng. DMFI: A dual-modality log analysis framework for insider threat detection with LoRA-tuned language models. *arXiv preprint arXiv:2508.05694*, 2025.

Sebastian Lapuschkin, Stephan Wäldchen, Alexander Binder, Grégoire Montavon, Wojciech Samek, and Klaus-Robert Müller. Unmasking clever hans predictors and assessing what machines really learn. *Nature Communications*, 10(1):1096, 2019.

Duc C. Le, Nur Zincir-Heywood, and Malcolm I. Heywood. Analyzing data granularity levels for insider threat detection using machine learning. *IEEE Transactions on Network and Service Management*, 17(1): 30–44, 2020.

Chenyu Li, Zhengjia Zhu, Jiyan He, and Xiu Zhang. Redchronos: A large language model-based log analysis system for insider threat detection in enterprises. *arXiv preprint arXiv:2503.02702*, 2025.

Brian Lindauer. Insider threat test dataset, 2020. Carnegie Mellon University, KiltHub repository. `https://kilthub.cmu.edu/articles/dataset/Insider_Threat_Test_Dataset/12841247`.

Fei Tony Liu, Kai Ming Ting, and Zhi-Hua Zhou. Isolation forest. In *IEEE International Conference on Data Mining*, 2008.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, 2022.

Chris Olah, Nick Cammarata, Ludwig Schubert, Gabriel Goh, Michael Petrov, and Shan Carter. Zoom in: An introduction to circuits. *Distill*, 2020.

K. Prasanth. Feature geometry of lora adapters: A sparse autoencoder analysis of representational divergence in fine-tuned language models. *arXiv preprint arXiv:2605.28896*, 2026.

Qwen Team. Qwen3 technical report, 2025. arXiv preprint arXiv:2505.09388.

Lukas Ruff, Robert A. Vandermeulen, Nico Görnitz, Lucas Deecke, Shoaib Ahmed Siddiqui, Alexander Binder, Emmanuel Müller, and Marius Kloft. Deep one-class classification. In *International Conference on Machine Learning*, 2018.

Aaron Tuor, Samuel Kaplan, Brian Hutchinson, Nicole Nichols, and Sean Robinson. Deep learning for unsupervised insider threat detection in structured cybersecurity data streams. *AAAI Workshop on Artificial Intelligence for Cyber Security*, 2017.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017.

Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, Sharon Qian, Daniel Nevo, Yaron Singer, and Stuart Shieber. Investigating gender bias in language models using causal mediation analysis. In *Advances in Neural Information Processing Systems*, 2020.

Kevin Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. Interpretability in the wild: A circuit for indirect object identification in GPT-2 small. In *International Conference on Learning Representations*, 2023.

Shuhan Yuan and Xintao Wu. Deep learning for insider threat detection: Review, challenges and opportunities. *Computers & Security*, 104, 2021.

Fred Zhang and Neel Nanda. Towards best practices of activation patching in language models: Metrics and methods, 2023. arXiv preprint arXiv:2309.16042.

A Methods Appendix

A.1 Serialization schema

Each user-day is rendered as: line 1, `DAY` with `week, project, role, b_unit, f_unit, dept, team, itadmin`; line 2, `PSY` with the five simulated trait values `O,C,E,A,N`; line 3, a session-count line; then one `SES` line per session with `key=value` pairs over the retained LC-DAL session fields: `pc`, work-hour/after-hour/weekend flags, `duration`, concurrency, start/end descriptors, activity counts (`n_allact, n_logon, n_usb, usb_mean_usb_dur, n_file, file_n_exef, n_email` with received/sent/attachment statistics, `n_http` with job/cloud/leak/hack-site counts). The four dropped fields are `file_n-to_usb1, file_n-from_usb1, file_n-file_act3, file_n-disk1` (hyphenated names are renamed by `itertuples(...)._asdict()` and were consequently omitted; the fast serializer reproduces this byte-faithfully).

A.2 Adaptation and scoring

QLoRA adaptation follows Section 4 exactly (NF4 double quantization, bf16 compute; LoRA r=16, α=32, dropout 0.05 on q/k/v/o/up/down/gate projections; sequence length 2048; effective batch 8; one epoch; lr 1.5×10^{-4}, cosine, 3% warmup). The anomaly score of example x is the attention-masked mean token NLL under the adapted model. When a batch's full fp32 logits would exceed 28 GiB, scoring switches to a chunked path that computes cross-entropy from hidden states via the output embedding in bounded slices; both paths are exact.

A.3 Token delta extraction and SAE training

Token deltas $\delta_t^{(\ell)} = h_t^{\text{ada}} - h_t^{\text{base}}$ are extracted at the residual stream after layer ℓ for every attention-valid token of every example in the evaluation pool (`eval.jsonl` order defines `example_idx`). SAEs are trained on rows standardized by per-dimension mean and standard deviation ($\tilde{\delta} = (\delta - \mu)/\sigma$, statistics stored with the checkpoint): encoder $z = \text{TopK}_k(\text{ReLU}(W_e\tilde{\delta} + b_e))$ with untied linear decoder $\hat{\tilde{\delta}} = W_d z$; Adam, lr 10^{-3}, 20 epochs, batch 1024–4096, at most 2×10^6 training rows per layer (all positive-example rows retained, benign rows subsampled). Search grid: layers $\{18, 26, 34\}$ (r6.2) and $\{18, 26\}$ (r4.2) $\times$ $m \in \{2, 4\}$ $\times$ $k \in \{4, 8\}$. Headline configurations—r6.2: ℓ=18, m=4, k=8 (reconstruction MSE 0.054 on standardized rows, effective L_0 7.98, dead-feature rate 1.0%); r4.2: ℓ=26, m=2, k=4 (MSE 0.093, L_0 4.0, dead rate 1.4%)—were selected by the exploratory causal contrast on the full positive population, which is why held-out evaluation in the main text claims feature-level rather than configuration-level independence.

A.4 Feature ranking and feature sets

For each feature f, with post-top-k activations $z_{t,f}$ over a labeled row sample, the ranking statistic is $\text{gap}_f = \overline{z_{\cdot,f}}^{\,\text{pos}} - \overline{z_{\cdot,f}}^{\,\text{ben}}$ (mean over positive-example rows minus mean over benign rows). `top`n is the n highest-gap features. The active control `control`n`_active` comprises the n features with smallest |gap| among those with row activation frequency ≥ 0.002, excluding `top5`. Behavioral (SES-restricted) ranking recomputes gap_f on SES-class rows only and excludes features whose profile-token activation-mass share on positives exceeds 0.5.

A.5 Causal patching procedure

For receiver r with token deltas $\{\delta_t\}$ and codes $\{z_t\}$, feature set F (top or control), donor d, and strength $\alpha \in \{0.25, 0.5, 0.75, 1.0\}$: (1) active positions $T_F = \{t : \sum_{f\in F} z_{t,f} > 0\}$; (2) donor prototype p_f = mean of $z_{t,f}^d$ over donor tokens where the donor's F-features are active (fallback: mean over all donor tokens); (3) counterfactual codes $z'_{t,f} = (1-\alpha)\, z_{t,f} + \alpha\, p_f$ for $t \in T_F, f \in F$, all other coordinates unchanged; (4) the patched delta is $\delta'_t = \sigma \odot W_d z'_t + \mu$ at *every* token, and the additive shift $\delta'_t - \delta_t$ is injected into the layer-ℓ residual stream by a forward hook during adapted-model scoring. Because step (4) substitutes the SAE reconstruction at all tokens in *both* arms, the two arms share the same SAE-reconstructed baseline:

reconstruction substitution is controlled across arms (common-mode rather than exactly canceling, since the downstream computation is nonlinear); the arms differ only in the feature edit at their active positions. The recorded per-receiver outcome for each (feature set, donor type) is the best (most negative) NLL change over the candidate-donor pool and α grid: the estimand is *maximal achievable repair* under the permitted donor/strength search, not an average causal effect over donors; best-donor selection is applied identically to top and control arms. As the primary answer to dose-matching, we additionally constructed *norm-matched* active controls—features matched to the top-5 on activation frequency, mean active magnitude, and decoder norm while remaining gap-neutral—which closes the edit-size gap to 1.33× (r6.2) and 1.79× (r4.2), and reran the headline contexts. All four top-versus-matched-control contrasts remain positive: r4.2 causal +0.00168 (cluster 95% CI $[0.00105, 0.00228]$; individually significant) and necessity +0.0012 $[-0.0016, 0.0038]$; r6.2 necessity +0.0609 $[0.0074, 0.0843]$ (significant) and causal +0.0059 $[-0.0018, 0.0096]$ (descriptive; four clusters). Each benchmark therefore retains at least one individually significant estimand at matched dose. Best-donor selection is also examined for extreme-value artifacts: donor pools are asymmetric (benign donors capped at 16 per receiver, anomalous donors scarcer: median 1 on r6.2, mean 13 on r4.2), which gives the benign arm more selection opportunities; the asymmetry is common to top and control arms, and a complementary median-over-donors estimand (per-donor best-α, median across donors, no extreme-value selection) remains positive on both headline contexts (+0.0005 on r6.2, +0.0006 on r4.2). A perturbation-norm audit of the headline candidate pools shows top-set edits are larger than active-control edits (mean best-row Frobenius norms $\approx$ 71 versus 20 on r6.2; 112–124 versus 11–13 on r4.2), an expected property of gap-ranked features. The reported estimand, however, is a donor-type contrast *within* feature set, and within-set edit norms are matched across benign and anomalous donor arms (differences of 0.2% on r6.2 and $\leq$ 6% on r4.2), so the top-minus-control contrast of donor-type contrasts is not attributable to perturbation magnitude alone. The reconstruction-substitution component common to both arms has mean per-token norm $\approx$ 7 (r6.2) and $\approx$ 37 (r4.2). As a size-calibration sensitivity, we recomputed the estimand on per-unit-norm effects (each best-row NLL change divided by its edit norm before forming the donor-type contrast), a conservative rescaling that assumes linear response and therefore favors the smaller control edits. On r4.2, the point contrast remains positive in three of four contexts including the headline (team +0.0008 per unit norm), though individual-context cluster intervals include zero; on r6.2, the contrast does not survive the rescaling (3 clusters, descriptive). r6.2's patch-repair advantage should therefore be read as size-uncalibrated: its causal-relevance evidence rests on the donor-type contrast and the necessity estimand, while the profile-bound content claim rests on token attribution, which involves no intervention. The same norm audit applied to the ablation estimand shows the analogous pattern: top-set ablations on positive receivers are larger than active-control ablations (mean Frobenius norm $\approx$ 45 versus $\approx$ 21 on r6.2; 165–169 versus 13–15 on r4.2) and are additionally asymmetric across receiver classes, because top features barely activate on the matched benign receivers (r6.2: 0.2–0.8 mean active tokens versus 3.0 on positives; r4.2: $\approx$ 2 versus $\approx$ 4.4), so the (benign − positive) contrast for the top set is dominated by the positive-side edit. The near-null benign-side activation is itself consistent with the attribution result—the selected features are positive-specific—but it means ablation, like patching, is size-uncalibrated across feature sets; its protections are the receiver-class contrast within feature set and cross-context, cross-dictionary replication, not norm matching.

A.6 Necessity ablation

Identical machinery with $z'_{t,f} = (1-\alpha)\, z_{t,f}$ for $f \in F, t \in T_F$ (scaling toward zero; no donor), applied to each positive receiver and its context-matched benign receiver; the estimand contrasts (benign minus positive) ablation effects, top versus control, as in Section 5.2.

A.7 Donor pools and matching

Context modes define donor eligibility by equality on the serialized fields: `role`; `dept`×`role`; `project`×`role`; `team`; `dept` (r4.2 uses `team, role, dept, dept_role`). Donor pools are benign-labeled or anomalous-labeled examples matching the receiver's context key, with the receiver's own user excluded; at most 16 candidate donors per receiver are drawn uniformly at random (seeded). Complete-case rates for the paired estimand are reported per context in the results artifacts (e.g., r4.2 causal: 1034–1157 complete of 1039–1301 available receivers; r6.2 `team` has no complete anomalous-donor case and is reported as unavailable). Benign donors and necessity matches are drawn from the evaluation pool, whose benign members are validation-split users never seen in training.

A.8 Detector protocols

Fold-aligned: seed 42; one held-out malicious user per fold; 800 benign test users per fold sampled from all benign users; day-level metrics over all test rows and user-level metrics after per-user max aggregation; the LM is trained once (its corpus covers $\sim$90% of benign users), baselines retrain per fold. User-disjoint: the benign comparison population is restricted to validation-split users (never trained on); r6.2: 405 of 4000 benign users, r4.2: 79 of 1000.

A.9 Uncertainty procedures

Primary: user-level cluster bootstrap, 10,000 draws resampling malicious users with replacement and re-computing the pooled complete-case contrast; supplementary: receiver/pair-level bootstrap (4,000 draws) and mean-of-user-means cluster bootstrap. Point estimates always equal the complete-case paired contrast. With four r6.2 clusters, percentile intervals are reported as descriptive.

A.10 Configuration-selection status

Data-dependent choices made on the full positive populations: layer, m, k, context-mode menu, feature-set sizes, and the α grid. Choices re-made inside discovery sets for held-out evaluation: feature identities (top and control sets). This is the feature-level independence claimed in the main text; configuration-level independence would require repeating the architecture/layer search within discovery users, which we leave to future work.

A.11 Compute

GPU experiments ran on A100-40GB nodes (Slurm) and two RTX 3070 (8 GB) cards; total GPU consumption for all mechanistic, confirmation, control, and ablation experiments was approximately 31 A100-hours plus $\sim$60 consumer-GPU-hours. Every job's batch configuration was validated by short debug-queue probes measuring worst-case peak memory before long runs.